

Week 3 evolutionary algorithms 1 questions

1. What are the stages in evolutionary algorithms
2. What changes can we do to the parent selection to make changes in the children
3. What are the competing forces in EA?
4. What is locus, allele, gene, phenotype and genotype?
5. How can we generate population?
6. Where in the cycle do we do selection?
7. What types of variation operators do we have?
8. What is mutation? Is it explorative or exploitative?
9. What is recombination? Is it explorative or exploitative?
10. What types of representations of genomes do we have in EA?
11. What is 1-point crossover, uniform crossover?
12. Permutation mutation: what is swap mutation, insert mutation, scramble mutation, inversion mutation?
13. Permutation recombination: what is partially mapped crossover, edge recombination, cycle crossover, order crossover?